Advanced Statistics for Physics Analysis

Project: ARPAV Temperature Trends in the Veneto Region

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1. Initialization and Data Loading Functions

2. Data Loading and Exploratory Data Analysis

2.1 Import Dataset

```
data_oderzo <- find_data("data/Oderzo.csv")
data_oderzo
```

```
## [[1]]
## # A tibble: 31 x 14
##     Anno   GEN   FEB   MAR   APR   MAG   GIU   LUG   AGO   SET   OTT   NOV   DIC
##    <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
##  1  1994   0.2  -1     4.6   6.1  11.3  13.4  16.7  17.3  13.1   5.5   5.2  -0.2
##  2  1995  -3.6  -0.3   0.9   4.2  10    13.1  17.1  14.3  10.2   6.2   0.8   0.8
##  3  1996   0.4  -3    -0.3   6.6  11.2  15    15    15.5  10.4   8     5.4  -0.1
##  4  1997   0.6  -0.2   2.2   3.8  11.1  15.2  15.4  16.3  11.7   6.9   4.4   1.6
##  5  1998   0.1  -1.2   0.6   7.2  11.6  15.7  16.3  16.2  11.9   7.4   0.6  -3.2
##  6  1999  -2.5  -3.6   2.8   7.2  13.3  14.6  16.6  17    13.5   9     2.3  -2.4
##  7  2000  -4.8  -1.7   2.2   8.4  12.4  15    14.5  16.7  11.8   9.6   5.6   2.3
##  8  2001   1.5  -0.9   5.7   5.5  13.2  13.4  16.6  16.4   9.8  11.4   1.1  -5.4
##  9  2002  -5.3   1.3   3.3   6.6  12.2  16.6  16.7  16.4  12.4   8.6   6.4   2.2
## 10  2003  -2.3  -4.7   1.3   6.6  12.9  19    18.4  19.6  11.4   6.2   5.5   0.2
## # i 21 more rows
## # i 1 more variable: Medio <dbl>
##
## [[2]]
## # A tibble: 31 x 14
##     Anno   GEN   FEB   MAR   APR   MAG   GIU   LUG   AGO   SET   OTT   NOV   DIC
##    <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
##  1  1994   4.4   3.6  10.3  11.4  16.9  19.3  24.3  24.2  18.2  11.2   8.7   3.4
##  2  1995   1.1   4.4   6.3  10.5  15.6  18.3  23.7  20.4  15.5  12.4   5.7   3.9
##  3  1996   3.4   1.7   5.5  12.3  17.2  21.8  21.5  21.4  15.7  12.5   9     3.3
##  4  1997   4.4   4.5   9.3  10.5  17.6  20.5  21.9  22.1  18.7  12.3   8.1   4.7
##  5  1998   3.8   5.3   7.5  11.9  17.8  21.9  23    23.9  17.5  12.3   5.5   1.1
##  6  1999   2.1   2.2   8.4  12.8  18.5  21    23.1  22.4  19.7  13.5   6.3   1.7
##  7  2000  -0.1   3.8   7.9  14.1  18.7  22.3  21.1  23.8  18    13.9   9.3   5.6
```

```
##  8  2001   4.8   4.6   9.8  11.4  19.4  20    22.6  23.5  15.5  14.9   5.9  -0.2
##  9  2002   0     5.2   9.8  12.6  17.8  22.8  23    22.1  17.5  13.2  10.2   5.3
## 10  2003   1.7   1.1   8.1  11.8  20.1  25.6  25.3  26.8  17.7  11     9     4.1
## # i 21 more rows
## # i 1 more variable: Medio <dbl>
##
## [[3]]
## # A tibble: 31 x 14
##     Anno   GEN   FEB   MAR   APR   MAG   GIU   LUG   AGO   SET   OTT   NOV   DIC
##    <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
##  1  1994   9.5   8.6  16.9  17    22.8  25.7  31.9  31.9  25.2  18.2  13.2   7.9
##  2  1995   6.7   9.7  12.2  16.8  21.3  24.3  31    28    22.4  21.1  11.5   7.4
##  3  1996   7.1   7.2  11.6  18.8  23.6  29    28.3  28.6  22.2  18.3  13.2   7.5
##  4  1997   9.1  10.6  17.1  17.4  24.4  26.6  29.4  29.5  27.5  18.6  12.6   8.5
##  5  1998   8    13.9  14.8  17.6  24.6  28.6  30.4  32.4  24.8  19    11.8   6.9
##  6  1999   8.6   9.1  14.5  18.7  24.3  27.9  30.6  29.4  27.9  19.7  12     6.9
##  7  2000   6.9  10.8  14.5  20.5  26    30.5  28.8  32.7  26.1  19.6  13.8   9.5
##  8  2001   8.2  11.7  14.5  18.2  26.7  27.1  29.9  31.7  22.5  19.5  12.7   6.7
##  9  2002   7.1   9.7  17.3  18.8  24.1  30.1  30.8  29.8  24.7  19.5  14.3   8.7
## 10  2003   7.2   8.7  16.2  17.7  27.8  33    32.7  35.4  26.1  16.5  13.5   8.8
## # i 21 more rows
## # i 1 more variable: Medio <dbl>
```

```
data_montesummano <- find_data("data/Monte_Summano.csv")
data_montesummano
```

```
## [[1]]
## # A tibble: 31 x 14
##     Anno   GEN   FEB   MAR   APR   MAG   GIU   LUG   AGO   SET   OTT   NOV   DIC
##    <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
##  1  1994  -0.4  -1.7   4.8   4.6   9    12.4  15.9  15.8  11.3   6     5     0.2
##  2  1995  -2.7   0.8   0.3   4.3   8    10.1  15.4  13     9.1   8.3   2.4   0.1
##  3  1996   0.1  -2.9   0     6.3   8.9  12.7  12.6  13.7   8.7   7.2   4.1   0.6
##  4  1997   0.1   0.1   3.4   1.9   8.8  11.9  12.9  14.1  11.2   5.4   3    -0.2
##  5  1998  -1.7   1.5   0.3   4.1   8.7  12.5  14    15.2  10.1   6.1   0.1  -1.9
##  6  1999  -0.6  -2.4   1.9   5.4  10.5  12.1  14.6  14.8  12.4   7.9   1.8  -2
##  7  2000  -3    -0.2   1.3   6.7  10.8  13.8  12.6  15.7  11.6   7.8   4.1   0.7
##  8  2001  -0.5   0     3.7   3.9  11.2  11.4  15    16.3   8.7   9.9   1.7  -3.7
##  9  2002  -2.1   0.6   3.5   4.8   9.8  13.9  14.2  14.3  10.3   7.6   5.1   0.7
## 10  2003   0.8  -2.8   3.7   5.6  12.8  18.1  17.5  20.3  12.5   6.3   5.3   1.1
## # i 21 more rows
## # i 1 more variable: Medio <dbl>
##
## [[2]]
## # A tibble: 31 x 14
##     Anno   GEN   FEB   MAR   APR   MAG   GIU   LUG   AGO   SET   OTT   NOV   DIC
##    <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
##  1  1994   3     1.8   8.8   8.4  13.4  17.3  21.5  21.5  15.1  10     7.9   2.9
##  2  1995   0.8   3.9   4.6   8.9  12.5  14.6  20.7  18    13.5  12.8   6     2.8
##  3  1996   2.7   0.7   3.7  10.4  13.8  18.2  17.8  18.6  13.2  10.5   6.8   3.4
##  4  1997   3.1   4     8.5   8.2  14    16.1  18.4  18.9  16.3   9.7   5.6   2.7
##  5  1998   1.4   5.9   5.4   7.7  13.7  17.5  19.9  21    14.4  10     3.7   1.5
##  6  1999   2.8   1.5   6     9.7  14.5  17.1  20    19.1  16.9  11.1   5     1.1
##  7  2000   0.9   3.9   5.6  11.2  15.6  19.6  18.2  21.3  16.3  10.8   6.7   3.5
```

```
##  8  2001   2.1   3.9   6.8   8.5  16.2  17.2  20.3  21.7  13.1  13.5   5.2   0.2
##  9  2002   1.1   3.9   8.3   9.1  13.8  18.9  19.2  18.7  14.4  11.1   7.8   3.2
## 10  2003   4.2   1.7   8.3   9.9  18.4  23.7  23.1  25.9  17.2   9.6   7.8   3.9
## # i 21 more rows
## # i 1 more variable: Medio <dbl>
##
## [[3]]
## # A tibble: 31 x 14
##     Anno   GEN   FEB   MAR   APR   MAG   GIU   LUG   AGO   SET   OTT   NOV   DIC
##    <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
##  1  1994   8.3   6.6  14.1  13.5  18.7  22.9  27.9  28.6  20.6  15.7  12     7.1
##  2  1995   6.5   8.2  10.8  14.4  17.8  19.9  27.1  24.8  19.5  19.9  10.8   6.7
##  3  1996   6.2   6.1   8.9  15.9  19.6  24.7  24.1  25.3  19.5  15.4  11.1   8
##  4  1997   7.9   9.9  15.2  16    20.2  21.4  25.2  25.6  24.1  16.5   9.7   6.7
##  5  1998   6.1  12.5  12.2  12.9  20.2  23.6  26.6  28.6  20.8  15.5   9.6   6.8
##  6  1999   8.3   7.7  11.3  15.6  19.5  23.6  26.6  25.1  23.7  16.6   9.9   5.9
##  7  2000   7.6  10.4  11.9  17    22.5  26.6  25.2  29.1  23.8  15.4  10.9   7.9
##  8  2001   6.2   9.9  11.2  14.7  22.5  23.8  27    29    19.6  19.8  11.5   6.2
##  9  2002   7.1   8.8  15.1  14.9  19.1  25    26.4  25.5  21.1  17.1  11.8   7.1
## 10  2003   9.9   9.2  15.4  16    26    31.4  30.4  33.7  24.6  14.5  11.9   7.8
## # i 21 more rows
## # i 1 more variable: Medio <dbl>
```

```
data_cittadella <- find_data("data/Cittadella.csv")
data_cittadella
```

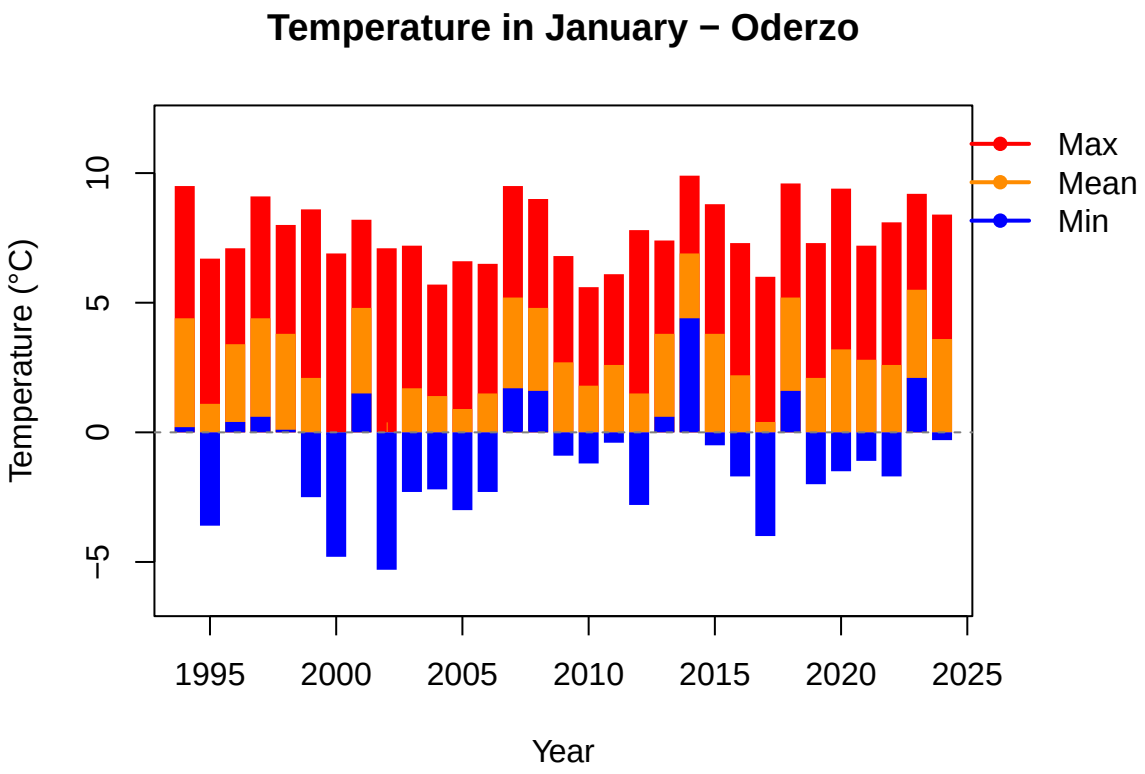
```
## [[1]]
## # A tibble: 31 x 14
##     Anno   GEN   FEB   MAR   APR   MAG   GIU   LUG   AGO   SET   OTT   NOV   DIC
##    <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
##  1  1994   0    -0.2   5.2   6.2  11.1  14.7  17.6  17    13.1   6.8   5.8   0.4
##  2  1995  -2.7   0.5   2.6   5.8  11    13.6  17.7  15    10.3   6.3   1.6   0.8
##  3  1996   1.3  -1.5   1.6   7.6  11.4  14.8  14.8  15.2  10.3   7.9   3.7  -1.9
##  4  1997   0.4  -0.4   2.8   4.5  11.6  15.1  15.4  16.8  12.7   6.9   4.4   2.2
##  5  1998   0.3  -0.1   1.2   7.1  11.3  15.1  16.5  16.3  11.7   7.1   0.4  -3.1
##  6  1999  -1.9  -2.8   3.3   7.6  13.4  14.3  16.5  16.5  13.4   8.6   1.9  -2.5
##  7  2000  -4.3  -1.4   2.7   9    12.3  14.5  14.5  15.9  11.9   9.9   5.3   1.6
##  8  2001   1.7   0.1   6     5.6  13.4  12.7  16.2  16.6   9.2  10.1   1.7  -5
##  9  2002  -5.1   1.1   4.2   7.1  11.9  15.8  16.4  15.8  12     8.3   6.2   2.3
## 10  2003  -0.3  -3.1   2.9   7.2  13.1  18    17.6  19.1  10.7   6.3   5.8   0.4
## # i 21 more rows
## # i 1 more variable: Medio <dbl>
##
## [[2]]
## # A tibble: 31 x 14
##     Anno   GEN   FEB   MAR   APR   MAG   GIU   LUG   AGO   SET   OTT   NOV   DIC
##    <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
##  1  1994   4.4   3.9  10.9  11.6  17.1  20.9  24.7  24    18.3  12.3   9.1   3.8
##  2  1995   2     5.1   7.7  11.8  16.5  19    24.5  21.1  16.3  12.8   6.7   4
##  3  1996   4.3   3     6.6  12.9  17.4  21.7  21.4  21.4  15.8  12.6   8     2.3
##  4  1997   4.3   4.9  10.5  11.7  18    20.5  22.5  22.9  19.8  12.7   8.3   5.4
##  5  1998   3.9   6.2   7.9  11.7  17.8  21.6  23.7  23.8  17.7  12.4   5.7   1.6
##  6  1999   2.6   2.8   8.7  13    18.5  21    23.2  22.6  19.8  13.4   6.1   1.9
##  7  2000   0.5   4.3   8.4  14.6  19.1  22.1  21.4  23.4  18.6  14.1   9     5.3
```

```
##  8  2001   4.7   5.5  10.2  11.8  20    20.4  23    24    15.8  15.4   6.7   0.2
##  9  2002   0.3   5.1  10.6  12.7  17.4  22.6  23.1  22.4  17.6  13.5  10.2   5.6
## 10  2003   3.1   2.4   9    12.1  20.2  24.7  24.4  26.4  17.6  11.3   9.3   4.6
## # i 21 more rows
## # i 1 more variable: Medio <dbl>
##
## [[3]]
## # A tibble: 31 x 14
##     Anno   GEN   FEB   MAR   APR   MAG   GIU   LUG   AGO   SET   OTT   NOV   DIC
##    <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
##  1  1994   9.4   8.6  17.2  16.8  22.6  27.2  31.6  31.7  25    18.8  13.4   8
##  2  1995   7.8  10.3  13.2  17.8  22.3  24.9  31.6  28.1  23.2  22.1  12.4   8.1
##  3  1996   7.8   8    11.8  18.4  23.5  28.4  28    28.2  22.3  18.2  13     7.3
##  4  1997   9.6  11.2  18.2  18.6  24.2  25.9  29.6  29.1  27.7  19.3  13     9
##  5  1998   8.3  14.4  14.6  16.9  24    28.5  31.1  32.2  25    18.8  11.9   7.9
##  6  1999   8.9   9.2  14.2  18.8  24.2  28    30.4  29.9  27.7  19.6  12     7.2
##  7  2000   7.5  10.8  14.6  20.5  26.1  29.6  28.9  32.2  26.7  19.6  14     9.7
##  8  2001   8    11.7  14.9  18    26.8  27.7  30.3  32.3  23.5  22.8  13.4   7.8
##  9  2002   7.8  10    17.4  18.5  23.6  30    31    30.2  25    19.9  14.7   9.1
## 10  2003   7.6   8.4  15.3  16.8  26.8  31.5  30.9  34.2  25.2  16.5  13.4   9.5
## # i 21 more rows
## # i 1 more variable: Medio <dbl>
```

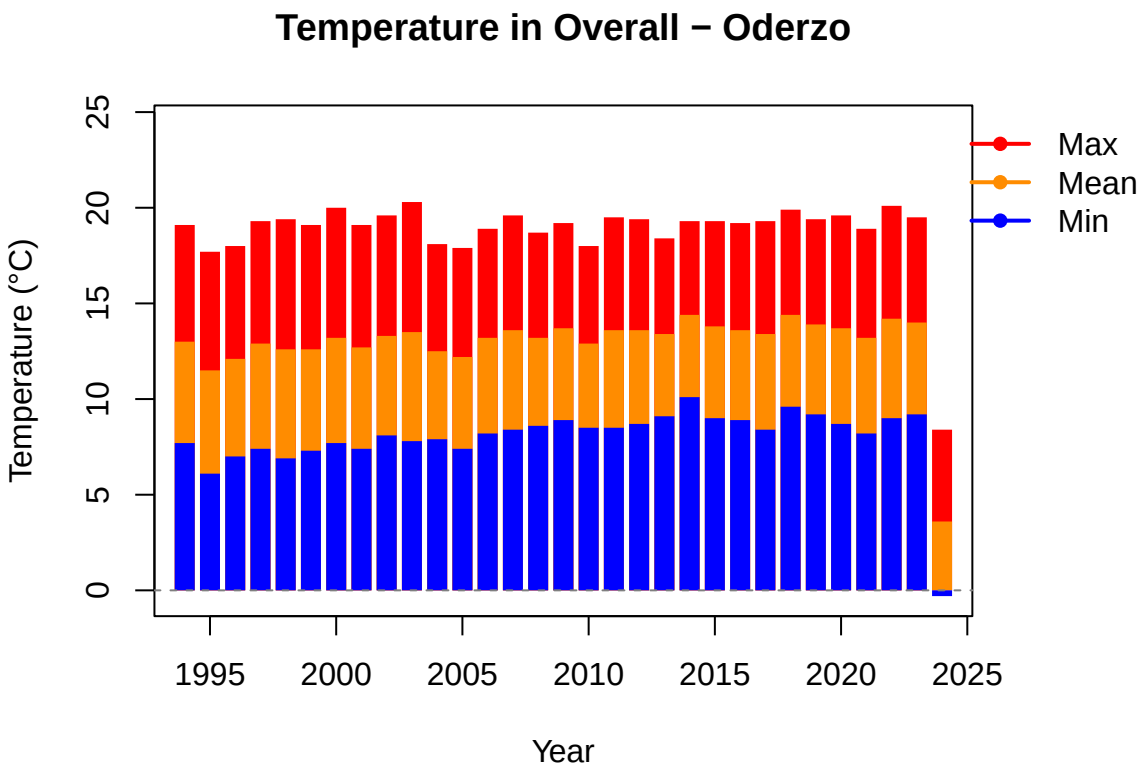
2.2 Exploratory Visualization (Oderzo)

```
temp_min <- data_oderzo[[1]]$GEN
temp_mean <- data_oderzo[[2]]$GEN
temp_max <- data_oderzo[[3]]$GEN

plot_hist(temp_min, temp_mean, temp_max, month="January", city="Oderzo")
```




```
plot_hist(data_oderzo[[1]]$Medio, data_oderzo[[2]]$Medio, data_oderzo[[3]]$Medio,
          month="Overall", city="Oderzo")
```



3. Bayesian Model Preparation (Monte Summano)

3.1 Summaries and Extraction

```
ms_means <- data_montesummano[[2]]
ms_means_yearavg <- ms_means$Medio

cat("Yearly means: \n")
```

```
## Yearly means:
```

```
cat(ms_means_yearavg, "\n")
```

```
## 11 9.9 10 10.5 10.2 10.4 11.1 10.7 10.8 12.8 10.8 10.3 11.3 12 11.3 11.8 10.3 11.9 11.7 11.3 12 12.2
```

```
ms_m_ya_length <- length(ms_means_yearavg)
cat("Number of yearly means: \n")
```

```
## Number of yearly means:
```

```r
cat(ms_m_ya_length, "\n")
```

```
## 31
```

```r
years <- ms_means$Anno
cat("Years: \n")
```

```
## Years:
```

```r
cat(years, "\n")
```

```
## 1994 1995 1996 1997 1998 1999 2000 2001 2002 2003 2004 2005 2006 2007 2008 2009 2010 2011 2012 2013
```

3.2 Standardization

```r
years_mean <- mean(years)
cat("Mean year:", years_mean, "\n")
```

```
## Mean year: 2009
```

```r
years_sd_ceiling <- ceiling(sd(years))
cat("Ceiling of standard deviation:", years_sd_ceiling, "\n")
```

```
## Ceiling of standard deviation: 10
```

```r
years_stdd <- (years - years_mean) / years_sd_ceiling

N <- ms_m_ya_length
cat("Number of observations:", N, "\n")
```

```
## Number of observations: 31
```

```r
step <- abs(years_stdd[1] - years_stdd[2])
cat("Standardized time step:", step, "\n")
```

```
## Standardized time step: 0.1
```

```r
N_fut_steps <- 10
cat("Number of future prediction steps:", N_fut_steps, "\n")
```

```
## Number of future prediction steps: 10
```

```r
time_future <- seq(1 + step, by = step, length.out = N_fut_steps)
cat("Future standardized time points:\n")
```

```
## Future standardized time points:
```

```r
print(time_future)
```

```
##  [1] 1.1 1.2 1.3 1.4 1.5 1.6 1.7 1.8 1.9 2.0
```

```r
stan_data_sdd <- list(
  N = N,
  time = years_stdd,
  T = ms_means_yearavg,
  N_future = N_fut_steps,
  time_future = time_future
)
```

4. Linear Model Fit (Stan)

4.1 MCMC Sampling and Convergence Diagnostics

```r
res <- build_fit_stan_model("./linear_model.stan", stan_data_sdd)
```

```
## MODEL COMPILED SUCCESSFULLY
## Running MCMC with 4 parallel chains...
##
## Chain 1 Iteration:    1 / 3000 [  0%]  (Warmup)
## Chain 1 Iteration:  100 / 3000 [  3%]  (Warmup)
## Chain 1 Iteration:  200 / 3000 [  6%]  (Warmup)
## Chain 1 Iteration:  300 / 3000 [ 10%]  (Warmup)
## Chain 1 Iteration:  400 / 3000 [ 13%]  (Warmup)
## Chain 1 Iteration:  500 / 3000 [ 16%]  (Warmup)
## Chain 1 Iteration:  600 / 3000 [ 20%]  (Warmup)
## Chain 1 Iteration:  700 / 3000 [ 23%]  (Warmup)
## Chain 1 Iteration:  800 / 3000 [ 26%]  (Warmup)
## Chain 1 Iteration:  900 / 3000 [ 30%]  (Warmup)
## Chain 1 Iteration: 1000 / 3000 [ 33%]  (Warmup)
## Chain 1 Iteration: 1001 / 3000 [ 33%]  (Sampling)
## Chain 1 Iteration: 1100 / 3000 [ 36%]  (Sampling)
## Chain 1 Iteration: 1200 / 3000 [ 40%]  (Sampling)
## Chain 1 Iteration: 1300 / 3000 [ 43%]  (Sampling)
## Chain 1 Iteration: 1400 / 3000 [ 46%]  (Sampling)
## Chain 1 Iteration: 1500 / 3000 [ 50%]  (Sampling)
## Chain 1 Iteration: 1600 / 3000 [ 53%]  (Sampling)
## Chain 1 Iteration: 1700 / 3000 [ 56%]  (Sampling)
## Chain 1 Iteration: 1800 / 3000 [ 60%]  (Sampling)
## Chain 2 Iteration:    1 / 3000 [  0%]  (Warmup)
## Chain 2 Iteration:  100 / 3000 [  3%]  (Warmup)
## Chain 2 Iteration:  200 / 3000 [  6%]  (Warmup)
## Chain 2 Iteration:  300 / 3000 [ 10%]  (Warmup)
## Chain 2 Iteration:  400 / 3000 [ 13%]  (Warmup)
## Chain 2 Iteration:  500 / 3000 [ 16%]  (Warmup)
## Chain 2 Iteration:  600 / 3000 [ 20%]  (Warmup)
## Chain 2 Iteration:  700 / 3000 [ 23%]  (Warmup)
```

```
## Chain 2 Iteration:  800 / 3000 [ 26%]  (Warmup)
## Chain 2 Iteration:  900 / 3000 [ 30%]  (Warmup)
## Chain 2 Iteration: 1000 / 3000 [ 33%]  (Warmup)
## Chain 2 Iteration: 1001 / 3000 [ 33%]  (Sampling)
## Chain 2 Iteration: 1100 / 3000 [ 36%]  (Sampling)
## Chain 2 Iteration: 1200 / 3000 [ 40%]  (Sampling)
## Chain 2 Iteration: 1300 / 3000 [ 43%]  (Sampling)
## Chain 2 Iteration: 1400 / 3000 [ 46%]  (Sampling)
## Chain 2 Iteration: 1500 / 3000 [ 50%]  (Sampling)
## Chain 2 Iteration: 1600 / 3000 [ 53%]  (Sampling)
## Chain 3 Iteration:    1 / 3000 [  0%]  (Warmup)
## Chain 3 Iteration:  100 / 3000 [  3%]  (Warmup)
## Chain 3 Iteration:  200 / 3000 [  6%]  (Warmup)
## Chain 3 Iteration:  300 / 3000 [ 10%]  (Warmup)
## Chain 3 Iteration:  400 / 3000 [ 13%]  (Warmup)
## Chain 3 Iteration:  500 / 3000 [ 16%]  (Warmup)
## Chain 3 Iteration:  600 / 3000 [ 20%]  (Warmup)
## Chain 3 Iteration:  700 / 3000 [ 23%]  (Warmup)
## Chain 3 Iteration:  800 / 3000 [ 26%]  (Warmup)
## Chain 3 Iteration:  900 / 3000 [ 30%]  (Warmup)
## Chain 3 Iteration: 1000 / 3000 [ 33%]  (Warmup)
## Chain 3 Iteration: 1001 / 3000 [ 33%]  (Sampling)
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## Chain 3 Iteration: 1500 / 3000 [ 50%]  (Sampling)
## Chain 4 Iteration:    1 / 3000 [  0%]  (Warmup)
## Chain 4 Iteration:  100 / 3000 [  3%]  (Warmup)
## Chain 4 Iteration:  200 / 3000 [  6%]  (Warmup)
## Chain 4 Iteration:  300 / 3000 [ 10%]  (Warmup)
## Chain 4 Iteration:  400 / 3000 [ 13%]  (Warmup)
## Chain 4 Iteration:  500 / 3000 [ 16%]  (Warmup)
## Chain 4 Iteration:  600 / 3000 [ 20%]  (Warmup)
## Chain 4 Iteration:  700 / 3000 [ 23%]  (Warmup)
## Chain 4 Iteration:  800 / 3000 [ 26%]  (Warmup)
## Chain 4 Iteration:  900 / 3000 [ 30%]  (Warmup)
## Chain 4 Iteration: 1000 / 3000 [ 33%]  (Warmup)
## Chain 4 Iteration: 1001 / 3000 [ 33%]  (Sampling)
## Chain 4 Iteration: 1100 / 3000 [ 36%]  (Sampling)
## Chain 4 Iteration: 1200 / 3000 [ 40%]  (Sampling)
## Chain 1 Iteration: 1900 / 3000 [ 63%]  (Sampling)
## Chain 1 Iteration: 2000 / 3000 [ 66%]  (Sampling)
## Chain 1 Iteration: 2100 / 3000 [ 70%]  (Sampling)
## Chain 1 Iteration: 2200 / 3000 [ 73%]  (Sampling)
## Chain 1 Iteration: 2300 / 3000 [ 76%]  (Sampling)
## Chain 1 Iteration: 2400 / 3000 [ 80%]  (Sampling)
## Chain 1 Iteration: 2500 / 3000 [ 83%]  (Sampling)
## Chain 1 Iteration: 2600 / 3000 [ 86%]  (Sampling)
## Chain 1 Iteration: 2700 / 3000 [ 90%]  (Sampling)
## Chain 1 Iteration: 2800 / 3000 [ 93%]  (Sampling)
## Chain 1 Iteration: 2900 / 3000 [ 96%]  (Sampling)
## Chain 1 Iteration: 3000 / 3000 [100%]  (Sampling)
## Chain 1 finished in 0.3 seconds.
```

```
## Chain 2 Iteration: 1700 / 3000 [ 56%]  (Sampling)
## Chain 2 Iteration: 1800 / 3000 [ 60%]  (Sampling)
## Chain 2 Iteration: 1900 / 3000 [ 63%]  (Sampling)
## Chain 2 Iteration: 2000 / 3000 [ 66%]  (Sampling)
## Chain 2 Iteration: 2100 / 3000 [ 70%]  (Sampling)
## Chain 2 Iteration: 2200 / 3000 [ 73%]  (Sampling)
## Chain 2 Iteration: 2300 / 3000 [ 76%]  (Sampling)
## Chain 2 Iteration: 2400 / 3000 [ 80%]  (Sampling)
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## Chain 2 Iteration: 2700 / 3000 [ 90%]  (Sampling)
## Chain 2 Iteration: 2800 / 3000 [ 93%]  (Sampling)
## Chain 2 Iteration: 2900 / 3000 [ 96%]  (Sampling)
## Chain 2 Iteration: 3000 / 3000 [100%]  (Sampling)
## Chain 2 finished in 0.3 seconds.
## Chain 3 Iteration: 1600 / 3000 [ 53%]  (Sampling)
## Chain 3 Iteration: 1700 / 3000 [ 56%]  (Sampling)
## Chain 3 Iteration: 1800 / 3000 [ 60%]  (Sampling)
## Chain 3 Iteration: 1900 / 3000 [ 63%]  (Sampling)
## Chain 3 Iteration: 2000 / 3000 [ 66%]  (Sampling)
## Chain 3 Iteration: 2100 / 3000 [ 70%]  (Sampling)
## Chain 3 Iteration: 2200 / 3000 [ 73%]  (Sampling)
## Chain 3 Iteration: 2300 / 3000 [ 76%]  (Sampling)
## Chain 3 Iteration: 2400 / 3000 [ 80%]  (Sampling)
## Chain 3 Iteration: 2500 / 3000 [ 83%]  (Sampling)
## Chain 3 Iteration: 2600 / 3000 [ 86%]  (Sampling)
## Chain 3 Iteration: 2700 / 3000 [ 90%]  (Sampling)
## Chain 3 Iteration: 2800 / 3000 [ 93%]  (Sampling)
## Chain 3 Iteration: 2900 / 3000 [ 96%]  (Sampling)
## Chain 3 Iteration: 3000 / 3000 [100%]  (Sampling)
## Chain 3 finished in 0.3 seconds.
## Chain 4 Iteration: 1300 / 3000 [ 43%]  (Sampling)
## Chain 4 Iteration: 1400 / 3000 [ 46%]  (Sampling)
## Chain 4 Iteration: 1500 / 3000 [ 50%]  (Sampling)
## Chain 4 Iteration: 1600 / 3000 [ 53%]  (Sampling)
## Chain 4 Iteration: 1700 / 3000 [ 56%]  (Sampling)
## Chain 4 Iteration: 1800 / 3000 [ 60%]  (Sampling)
## Chain 4 Iteration: 1900 / 3000 [ 63%]  (Sampling)
## Chain 4 Iteration: 2000 / 3000 [ 66%]  (Sampling)
## Chain 4 Iteration: 2100 / 3000 [ 70%]  (Sampling)
## Chain 4 Iteration: 2200 / 3000 [ 73%]  (Sampling)
## Chain 4 Iteration: 2300 / 3000 [ 76%]  (Sampling)
## Chain 4 Iteration: 2400 / 3000 [ 80%]  (Sampling)
## Chain 4 Iteration: 2500 / 3000 [ 83%]  (Sampling)
## Chain 4 Iteration: 2600 / 3000 [ 86%]  (Sampling)
## Chain 4 Iteration: 2700 / 3000 [ 90%]  (Sampling)
## Chain 4 Iteration: 2800 / 3000 [ 93%]  (Sampling)
## Chain 4 Iteration: 2900 / 3000 [ 96%]  (Sampling)
## Chain 4 Iteration: 3000 / 3000 [100%]  (Sampling)
## Chain 4 finished in 0.3 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 0.3 seconds.
## Total execution time: 0.9 seconds.
```

```
mod <- res$mod
fit <- res$fit

stan_T_param_names <- c("T0", "T_dot")

cat("Summary of T parameters:\n")
```

```
## Summary of T parameters:
```

```
fit$summary(stan_T_param_names)
```

```
## # A tibble: 2 x 10
##   variable   mean median    sd   mad     q5    q95  rhat ess_bulk ess_tail
##   <chr>     <dbl>  <dbl> <dbl> <dbl>  <dbl>  <dbl> <dbl>    <dbl>    <dbl>
## 1 T0       11.4   11.4   0.107 0.107 11.2   11.6    1.00    6240.    5499.
## 2 T_dot     0.720  0.721 0.121 0.118  0.523  0.919  1.00    7328.    5393.
```

```
cat("Summary of sigma:\n")
```

```
## Summary of sigma:
```

```
fit$summary("sigma")
```

```
## # A tibble: 1 x 10
##   variable  mean median     sd    mad    q5   q95  rhat ess_bulk ess_tail
##   <chr>    <dbl>  <dbl>  <dbl>  <dbl> <dbl> <dbl> <dbl>    <dbl>    <dbl>
## 1 sigma    0.601  0.593 0.0820 0.0769 0.482 0.749  1.00    7219.    5738.
```

```
cat("Running CmdStan diagnostics...\n")
```

```
## Running CmdStan diagnostics...
```

```
fit$cmdstan_diagnose()
```

```
## Checking sampler transitions treedepth.
## Treedepth satisfactory for all transitions.
##
## Checking sampler transitions for divergences.
## No divergent transitions found.
##
## Checking E-BFMI - sampler transitions HMC potential energy.
## E-BFMI satisfactory.
##
## Rank-normalized split effective sample size satisfactory for all parameters.
##
## Rank-normalized split R-hat values satisfactory for all parameters.
##
## Processing complete, no problems detected.
```

4.2 95% Credible Intervals

```
draws_df <- as_draws_df(fit$draws())

quantiles_T <- lapply(
  draws_df[, stan_T_param_names],
  quantile,
  probs = c(0.025, 0.975)
)

cat("T0    : [",
    round(quantiles_T$T0[1], 3), ", ",
    round(quantiles_T$T0[2], 3), "]\n")
```

```
## T0    : [ 11.186 ,  11.609 ]
```

```
cat("T_dot : [",
    round(quantiles_T$T_dot[1], 3), ", ",
    round(quantiles_T$T_dot[2], 3), "]\n")
```

```
## T_dot : [ 0.481 ,  0.958 ]
```

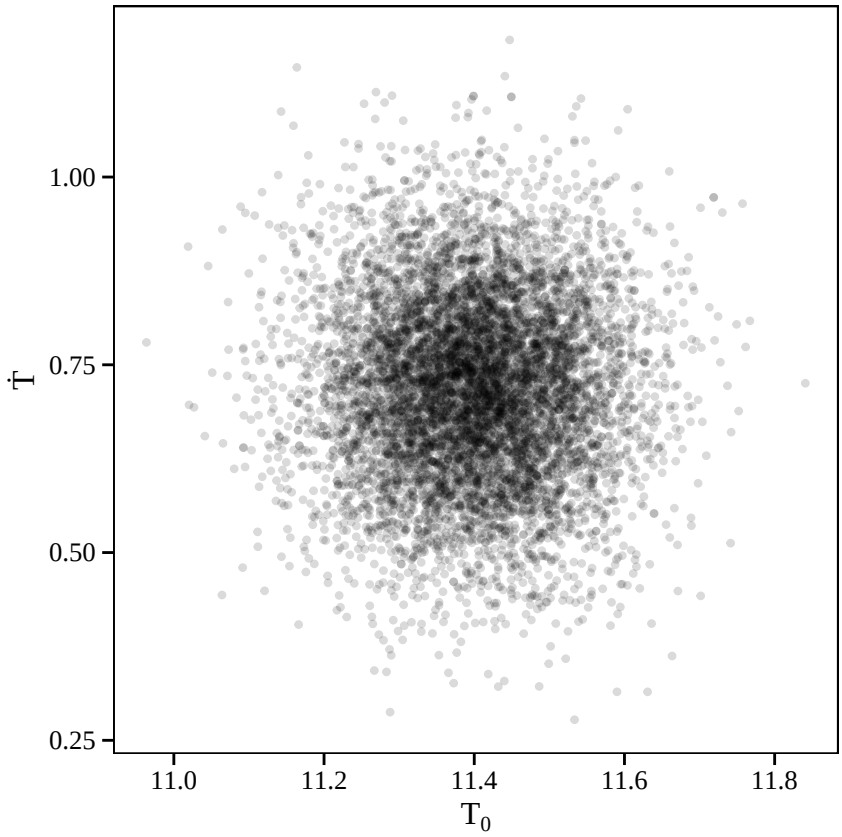
```
cat("\n")
```

4.3 Parameter Dependencies (Correlation Analysis)

```
cor_plot <- plot_2_correlations(draws_df, stan_T_param_names)
```

```
## Warning: Dropping 'draws_df' class as required metadata was removed.
```

```
print(cor_plot$plot)
```



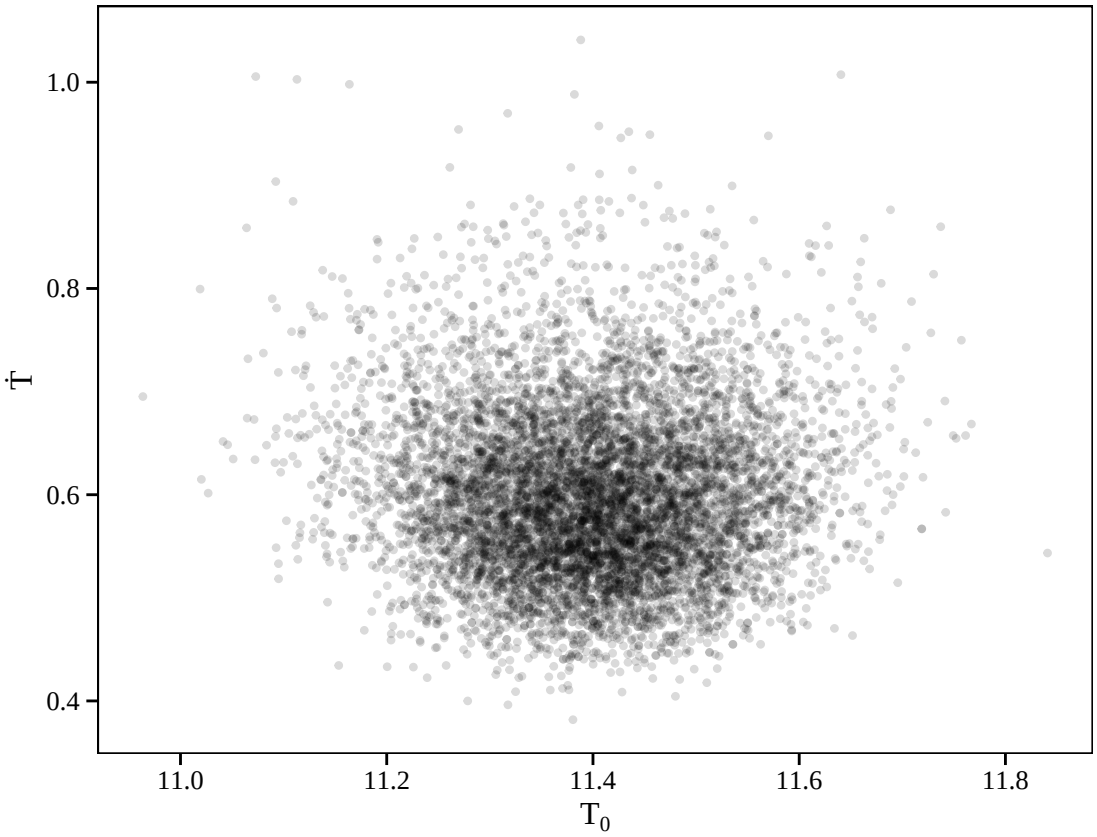

```
cor_plot$cor
```

```
##                T0       T_dot
## T0     1.00000000 -0.01344195
## T_dot -0.01344195  1.00000000
```

```
cor_plot_sigma_0 <- plot_2_correlations(draws_df, c("T0", "sigma"))
```

```
## Warning: Dropping 'draws_df' class as required metadata was removed.
```

```
print(cor_plot_sigma_0$plot)
```



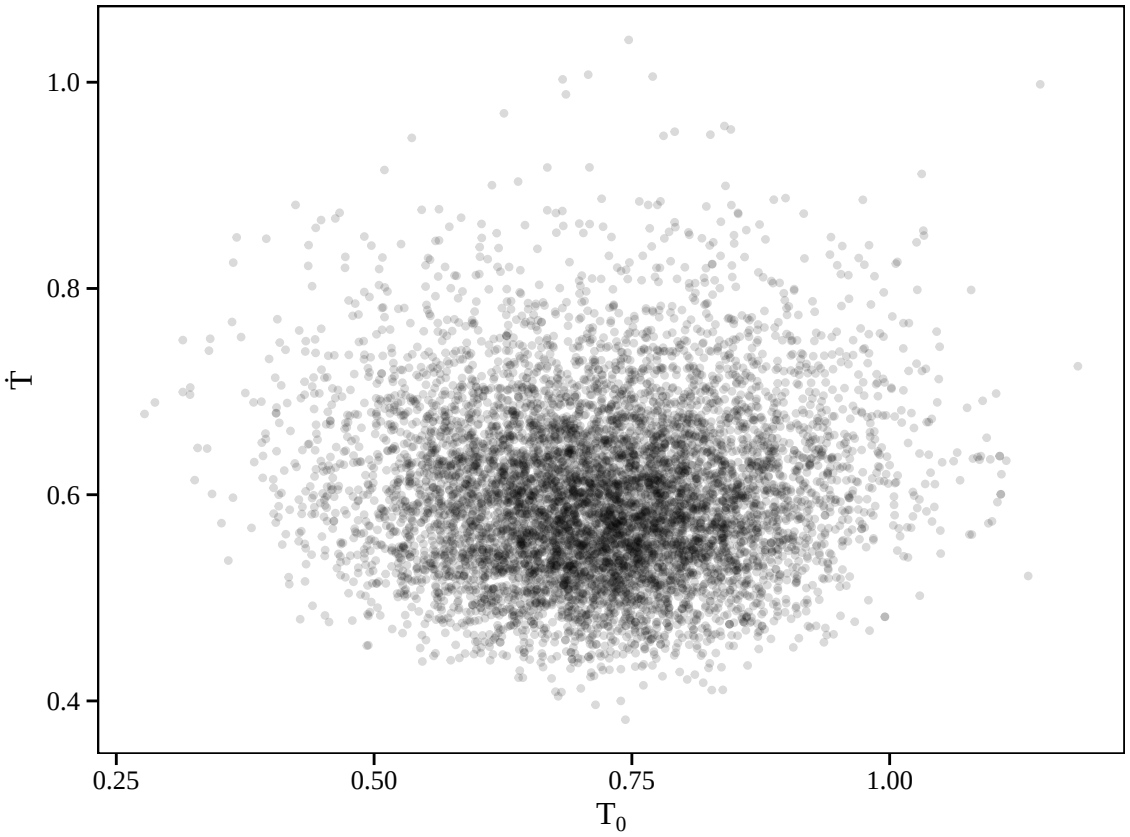

```
cor_plot_sigma_0$cor
```

```
##                T0       sigma
## T0     1.00000000 -0.01473961
## sigma -0.01473961  1.00000000
```

```
cor_plot_sigma_dot <- plot_2_correlations(draws_df, c("T_dot", "sigma"))
```

```
## Warning: Dropping 'draws_df' class as required metadata was removed.
```

```
print(cor_plot_sigma_dot$plot)
```

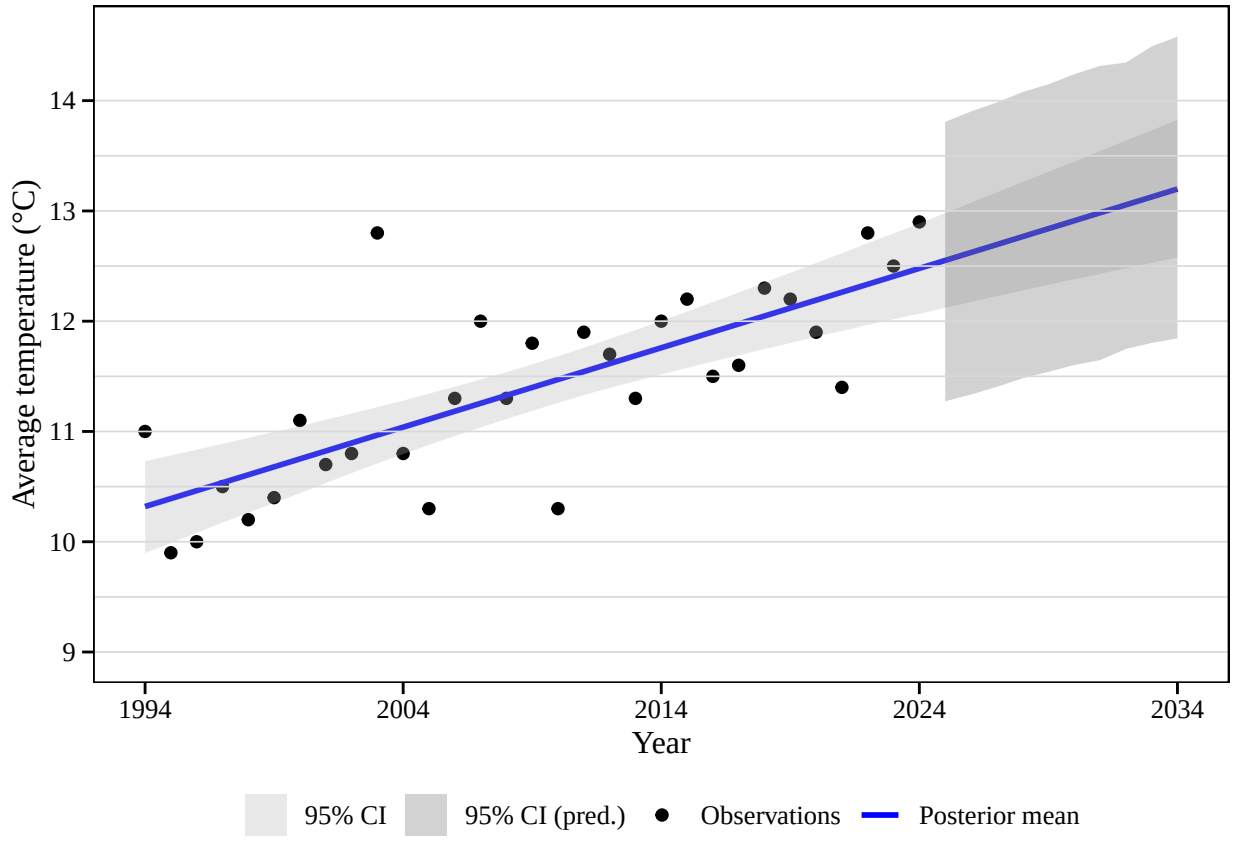



```
cor_plot_sigma_dot$cor
```

```
##             T_dot      sigma
## T_dot 1.00000000 0.01162347
## sigma 0.01162347 1.00000000
```

4.4 Posterior Density and Observational Fitting

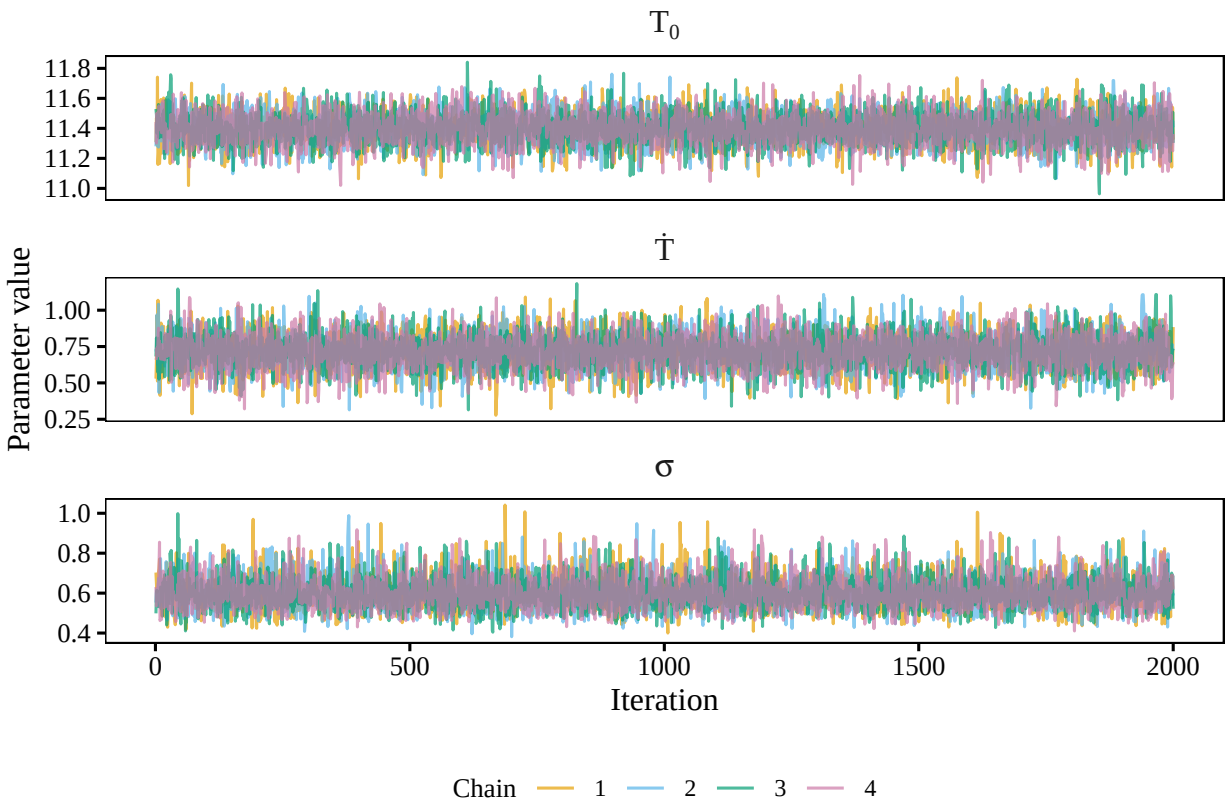
```
plot <- plot_posterior(draws_df, my_x_stdd = stan_data_sdd$time, my_y_data = stan_data_sdd$T)
print(plot)
```



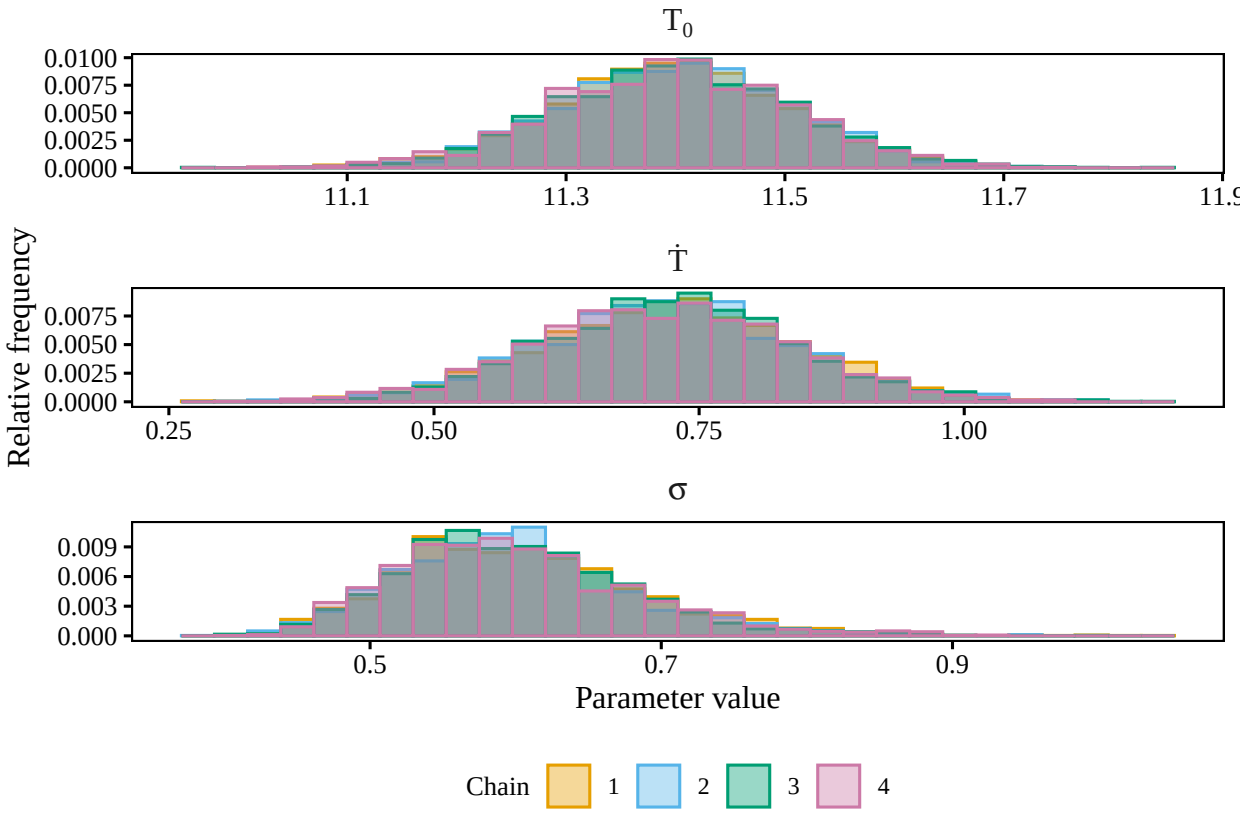
4.5 MCMC Diagnostic Plots

```
graphs <- plot_various_graphs(fit, c("T0", "T_dot", "sigma"))

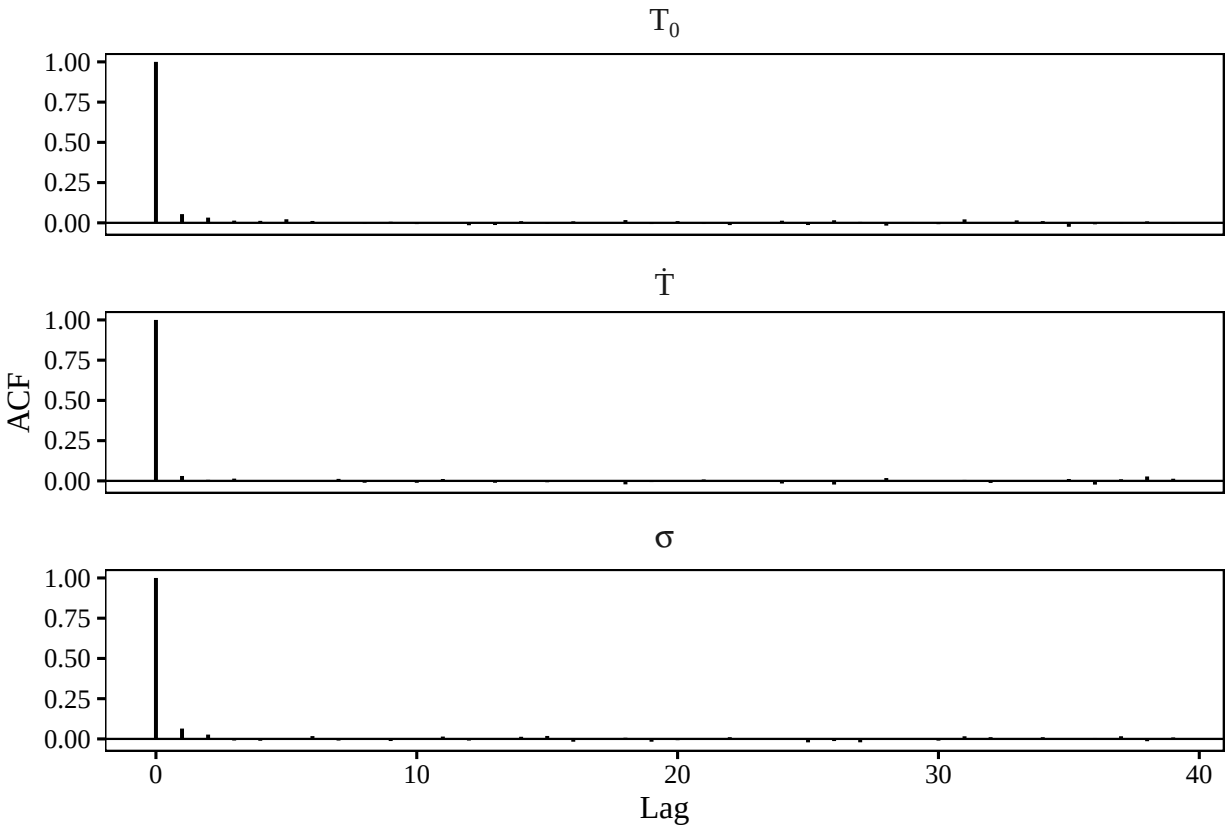
print(graphs$trace)
```




```
print(graphs$density)
```




```
print(graphs$acf)
```



5. Constant Model Fit (Stan)

5.1 MCMC Sampling and Convergence Diagnostics

```
res_c <- build_fit_stan_model("./constant_model.stan", stan_data_sdd)
```

```
## MODEL COMPILED SUCCESSFULLY
## Running MCMC with 4 parallel chains...
##
## Chain 1 Iteration:    1 / 3000 [  0%]  (Warmup)
## Chain 1 Iteration:  100 / 3000 [  3%]  (Warmup)
## Chain 1 Iteration:  200 / 3000 [  6%]  (Warmup)
## Chain 1 Iteration:  300 / 3000 [ 10%]  (Warmup)
## Chain 1 Iteration:  400 / 3000 [ 13%]  (Warmup)
## Chain 1 Iteration:  500 / 3000 [ 16%]  (Warmup)
## Chain 1 Iteration:  600 / 3000 [ 20%]  (Warmup)
## Chain 1 Iteration:  700 / 3000 [ 23%]  (Warmup)
## Chain 1 Iteration:  800 / 3000 [ 26%]  (Warmup)
## Chain 1 Iteration:  900 / 3000 [ 30%]  (Warmup)
## Chain 1 Iteration: 1000 / 3000 [ 33%]  (Warmup)
## Chain 1 Iteration: 1001 / 3000 [ 33%]  (Sampling)
## Chain 1 Iteration: 1100 / 3000 [ 36%]  (Sampling)
## Chain 1 Iteration: 1200 / 3000 [ 40%]  (Sampling)
```

```
## Chain 1 Iteration: 1300 / 3000 [ 43%]  (Sampling)
## Chain 1 Iteration: 1400 / 3000 [ 46%]  (Sampling)
## Chain 1 Iteration: 1500 / 3000 [ 50%]  (Sampling)
## Chain 1 Iteration: 1600 / 3000 [ 53%]  (Sampling)
## Chain 1 Iteration: 1700 / 3000 [ 56%]  (Sampling)
## Chain 2 Iteration:    1 / 3000 [  0%]  (Warmup)
## Chain 2 Iteration:  100 / 3000 [  3%]  (Warmup)
## Chain 2 Iteration:  200 / 3000 [  6%]  (Warmup)
## Chain 2 Iteration:  300 / 3000 [ 10%]  (Warmup)
## Chain 2 Iteration:  400 / 3000 [ 13%]  (Warmup)
## Chain 2 Iteration:  500 / 3000 [ 16%]  (Warmup)
## Chain 2 Iteration:  600 / 3000 [ 20%]  (Warmup)
## Chain 2 Iteration:  700 / 3000 [ 23%]  (Warmup)
## Chain 2 Iteration:  800 / 3000 [ 26%]  (Warmup)
## Chain 2 Iteration:  900 / 3000 [ 30%]  (Warmup)
## Chain 2 Iteration: 1000 / 3000 [ 33%]  (Warmup)
## Chain 2 Iteration: 1001 / 3000 [ 33%]  (Sampling)
## Chain 2 Iteration: 1100 / 3000 [ 36%]  (Sampling)
## Chain 2 Iteration: 1200 / 3000 [ 40%]  (Sampling)
## Chain 2 Iteration: 1300 / 3000 [ 43%]  (Sampling)
## Chain 2 Iteration: 1400 / 3000 [ 46%]  (Sampling)
## Chain 2 Iteration: 1500 / 3000 [ 50%]  (Sampling)
## Chain 2 Iteration: 1600 / 3000 [ 53%]  (Sampling)
## Chain 2 Iteration: 1700 / 3000 [ 56%]  (Sampling)
## Chain 3 Iteration:    1 / 3000 [  0%]  (Warmup)
## Chain 3 Iteration:  100 / 3000 [  3%]  (Warmup)
## Chain 3 Iteration:  200 / 3000 [  6%]  (Warmup)
## Chain 3 Iteration:  300 / 3000 [ 10%]  (Warmup)
## Chain 3 Iteration:  400 / 3000 [ 13%]  (Warmup)
## Chain 3 Iteration:  500 / 3000 [ 16%]  (Warmup)
## Chain 3 Iteration:  600 / 3000 [ 20%]  (Warmup)
## Chain 3 Iteration:  700 / 3000 [ 23%]  (Warmup)
## Chain 3 Iteration:  800 / 3000 [ 26%]  (Warmup)
## Chain 3 Iteration:  900 / 3000 [ 30%]  (Warmup)
## Chain 3 Iteration: 1000 / 3000 [ 33%]  (Warmup)
## Chain 3 Iteration: 1001 / 3000 [ 33%]  (Sampling)
## Chain 3 Iteration: 1100 / 3000 [ 36%]  (Sampling)
## Chain 3 Iteration: 1200 / 3000 [ 40%]  (Sampling)
## Chain 3 Iteration: 1300 / 3000 [ 43%]  (Sampling)
## Chain 3 Iteration: 1400 / 3000 [ 46%]  (Sampling)
## Chain 3 Iteration: 1500 / 3000 [ 50%]  (Sampling)
## Chain 3 Iteration: 1600 / 3000 [ 53%]  (Sampling)
## Chain 4 Iteration:    1 / 3000 [  0%]  (Warmup)
## Chain 4 Iteration:  100 / 3000 [  3%]  (Warmup)
## Chain 4 Iteration:  200 / 3000 [  6%]  (Warmup)
## Chain 4 Iteration:  300 / 3000 [ 10%]  (Warmup)
## Chain 4 Iteration:  400 / 3000 [ 13%]  (Warmup)
## Chain 4 Iteration:  500 / 3000 [ 16%]  (Warmup)
## Chain 4 Iteration:  600 / 3000 [ 20%]  (Warmup)
## Chain 4 Iteration:  700 / 3000 [ 23%]  (Warmup)
## Chain 4 Iteration:  800 / 3000 [ 26%]  (Warmup)
## Chain 4 Iteration:  900 / 3000 [ 30%]  (Warmup)
## Chain 4 Iteration: 1000 / 3000 [ 33%]  (Warmup)
## Chain 4 Iteration: 1001 / 3000 [ 33%]  (Sampling)
```

```
## Chain 4 Iteration: 1100 / 3000 [ 36%]  (Sampling)
## Chain 4 Iteration: 1200 / 3000 [ 40%]  (Sampling)
## Chain 4 Iteration: 1300 / 3000 [ 43%]  (Sampling)
## Chain 4 Iteration: 1400 / 3000 [ 46%]  (Sampling)
## Chain 1 Iteration: 1800 / 3000 [ 60%]  (Sampling)
## Chain 1 Iteration: 1900 / 3000 [ 63%]  (Sampling)
## Chain 1 Iteration: 2000 / 3000 [ 66%]  (Sampling)
## Chain 1 Iteration: 2100 / 3000 [ 70%]  (Sampling)
## Chain 1 Iteration: 2200 / 3000 [ 73%]  (Sampling)
## Chain 1 Iteration: 2300 / 3000 [ 76%]  (Sampling)
## Chain 1 Iteration: 2400 / 3000 [ 80%]  (Sampling)
## Chain 1 Iteration: 2500 / 3000 [ 83%]  (Sampling)
## Chain 1 Iteration: 2600 / 3000 [ 86%]  (Sampling)
## Chain 1 Iteration: 2700 / 3000 [ 90%]  (Sampling)
## Chain 1 Iteration: 2800 / 3000 [ 93%]  (Sampling)
## Chain 1 Iteration: 2900 / 3000 [ 96%]  (Sampling)
## Chain 1 Iteration: 3000 / 3000 [100%]  (Sampling)
## Chain 1 finished in 0.3 seconds.
## Chain 2 Iteration: 1800 / 3000 [ 60%]  (Sampling)
## Chain 2 Iteration: 1900 / 3000 [ 63%]  (Sampling)
## Chain 2 Iteration: 2000 / 3000 [ 66%]  (Sampling)
## Chain 2 Iteration: 2100 / 3000 [ 70%]  (Sampling)
## Chain 2 Iteration: 2200 / 3000 [ 73%]  (Sampling)
## Chain 2 Iteration: 2300 / 3000 [ 76%]  (Sampling)
## Chain 2 Iteration: 2400 / 3000 [ 80%]  (Sampling)
## Chain 2 Iteration: 2500 / 3000 [ 83%]  (Sampling)
## Chain 2 Iteration: 2600 / 3000 [ 86%]  (Sampling)
## Chain 2 Iteration: 2700 / 3000 [ 90%]  (Sampling)
## Chain 2 Iteration: 2800 / 3000 [ 93%]  (Sampling)
## Chain 2 Iteration: 2900 / 3000 [ 96%]  (Sampling)
## Chain 2 Iteration: 3000 / 3000 [100%]  (Sampling)
## Chain 2 finished in 0.3 seconds.
## Chain 3 Iteration: 1700 / 3000 [ 56%]  (Sampling)
## Chain 3 Iteration: 1800 / 3000 [ 60%]  (Sampling)
## Chain 3 Iteration: 1900 / 3000 [ 63%]  (Sampling)
## Chain 3 Iteration: 2000 / 3000 [ 66%]  (Sampling)
## Chain 3 Iteration: 2100 / 3000 [ 70%]  (Sampling)
## Chain 3 Iteration: 2200 / 3000 [ 73%]  (Sampling)
## Chain 3 Iteration: 2300 / 3000 [ 76%]  (Sampling)
## Chain 3 Iteration: 2400 / 3000 [ 80%]  (Sampling)
## Chain 3 Iteration: 2500 / 3000 [ 83%]  (Sampling)
## Chain 3 Iteration: 2600 / 3000 [ 86%]  (Sampling)
## Chain 3 Iteration: 2700 / 3000 [ 90%]  (Sampling)
## Chain 3 Iteration: 2800 / 3000 [ 93%]  (Sampling)
## Chain 3 Iteration: 2900 / 3000 [ 96%]  (Sampling)
## Chain 3 Iteration: 3000 / 3000 [100%]  (Sampling)
## Chain 3 finished in 0.2 seconds.
## Chain 4 Iteration: 1500 / 3000 [ 50%]  (Sampling)
## Chain 4 Iteration: 1600 / 3000 [ 53%]  (Sampling)
## Chain 4 Iteration: 1700 / 3000 [ 56%]  (Sampling)
## Chain 4 Iteration: 1800 / 3000 [ 60%]  (Sampling)
## Chain 4 Iteration: 1900 / 3000 [ 63%]  (Sampling)
## Chain 4 Iteration: 2000 / 3000 [ 66%]  (Sampling)
## Chain 4 Iteration: 2100 / 3000 [ 70%]  (Sampling)
```

```
## Chain 4 Iteration: 2200 / 3000 [ 73%]  (Sampling)
## Chain 4 Iteration: 2300 / 3000 [ 76%]  (Sampling)
## Chain 4 Iteration: 2400 / 3000 [ 80%]  (Sampling)
## Chain 4 Iteration: 2500 / 3000 [ 83%]  (Sampling)
## Chain 4 Iteration: 2600 / 3000 [ 86%]  (Sampling)
## Chain 4 Iteration: 2700 / 3000 [ 90%]  (Sampling)
## Chain 4 Iteration: 2800 / 3000 [ 93%]  (Sampling)
## Chain 4 Iteration: 2900 / 3000 [ 96%]  (Sampling)
## Chain 4 Iteration: 3000 / 3000 [100%]  (Sampling)
## Chain 4 finished in 0.3 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 0.3 seconds.
## Total execution time: 0.4 seconds.
```

```
mod_c <- res_c$mod
fit_c <- res_c$fit

cat("Summary of T, sigma parameters:\n")
```

```
## Summary of T, sigma parameters:
```

```
fit_c$summary(c("T0", "sigma"))
```

```
## # A tibble: 2 x 10
##   variable   mean median    sd   mad     q5   q95  rhat ess_bulk ess_tail
##   <chr>     <dbl>  <dbl> <dbl> <dbl>  <dbl> <dbl> <dbl>    <dbl>    <dbl>
## 1 T0       11.4   11.4   0.163 0.157 11.1   11.7   1.00    5772.    4888.
## 2 sigma     0.902  0.890 0.122 0.116  0.726  1.12  1.00    6791.    5244.
```

```
cat("Running CmdStan diagnostics...\n")
```

```
## Running CmdStan diagnostics...
```

```
fit_c$cmdstan_diagnose()
```

```
## Checking sampler transitions treedepth.
## Treedepth satisfactory for all transitions.
##
## Checking sampler transitions for divergences.
## No divergent transitions found.
##
## Checking E-BFMI - sampler transitions HMC potential energy.
## E-BFMI satisfactory.
##
## Rank-normalized split effective sample size satisfactory for all parameters.
##
## Rank-normalized split R-hat values satisfactory for all parameters.
##
## Processing complete, no problems detected.
```

5.2 95% Credible Intervals

```
draws_df_c <- as_draws_df(fit_c$draws())

quantiles_c <- lapply(
  draws_df_c[, c("T0", "sigma")],
  quantile,
  probs = c(0.025, 0.975)
)

cat("T0 : [",
    round(quantiles_c$T0[1], 3), ", ",
    round(quantiles_c$T0[2], 3), "]\n")
```

```
## T0 : [ 11.075 ,  11.728 ]
```

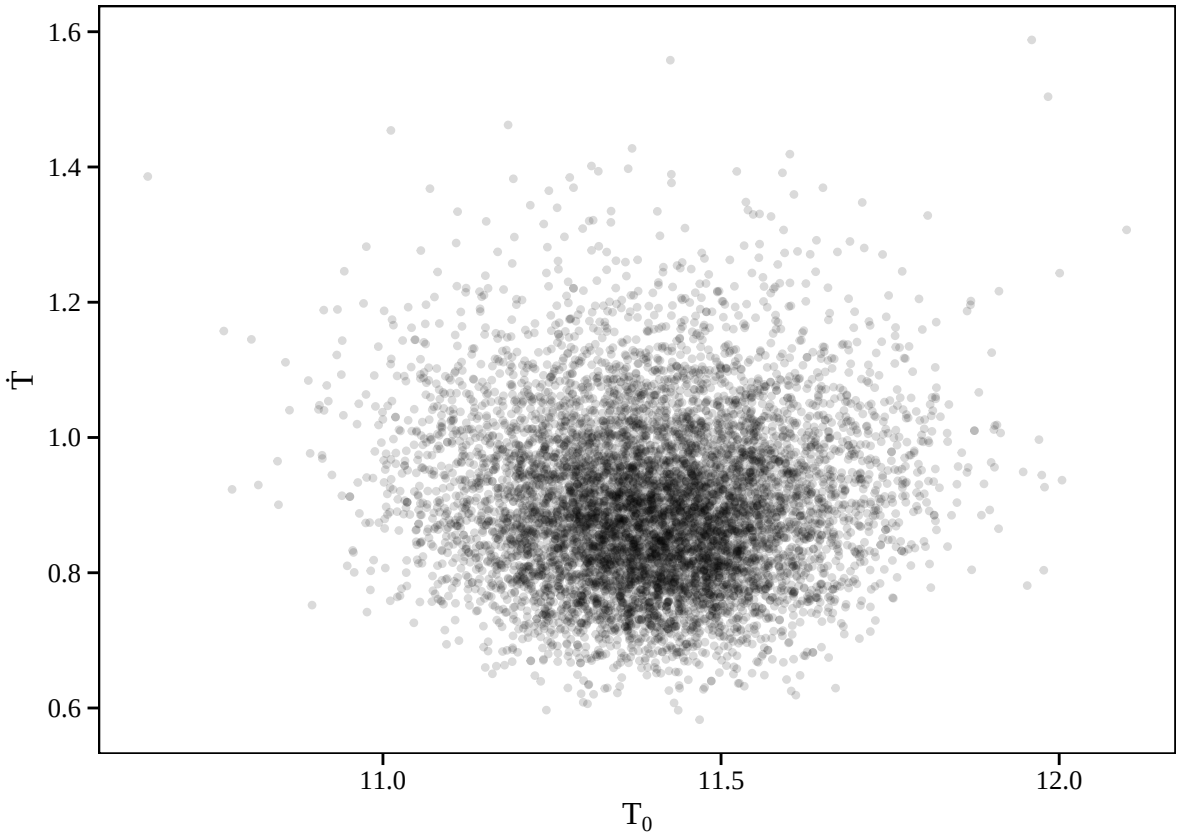
```
cat("sigma : [",
    round(quantiles_c$sigma[1], 3), ", ",
    round(quantiles_c$sigma[2], 3), "]\n")
```

```
## sigma : [ 0.698 ,  1.177 ]
```

```
cat("\n")
```

5.3 Parameter Dependencies (Correlation Analysis)

```
cor_plot_c <- plot_2_correlations(draws_df_c, c("T0", "sigma"))
print(cor_plot_c$plot)
```

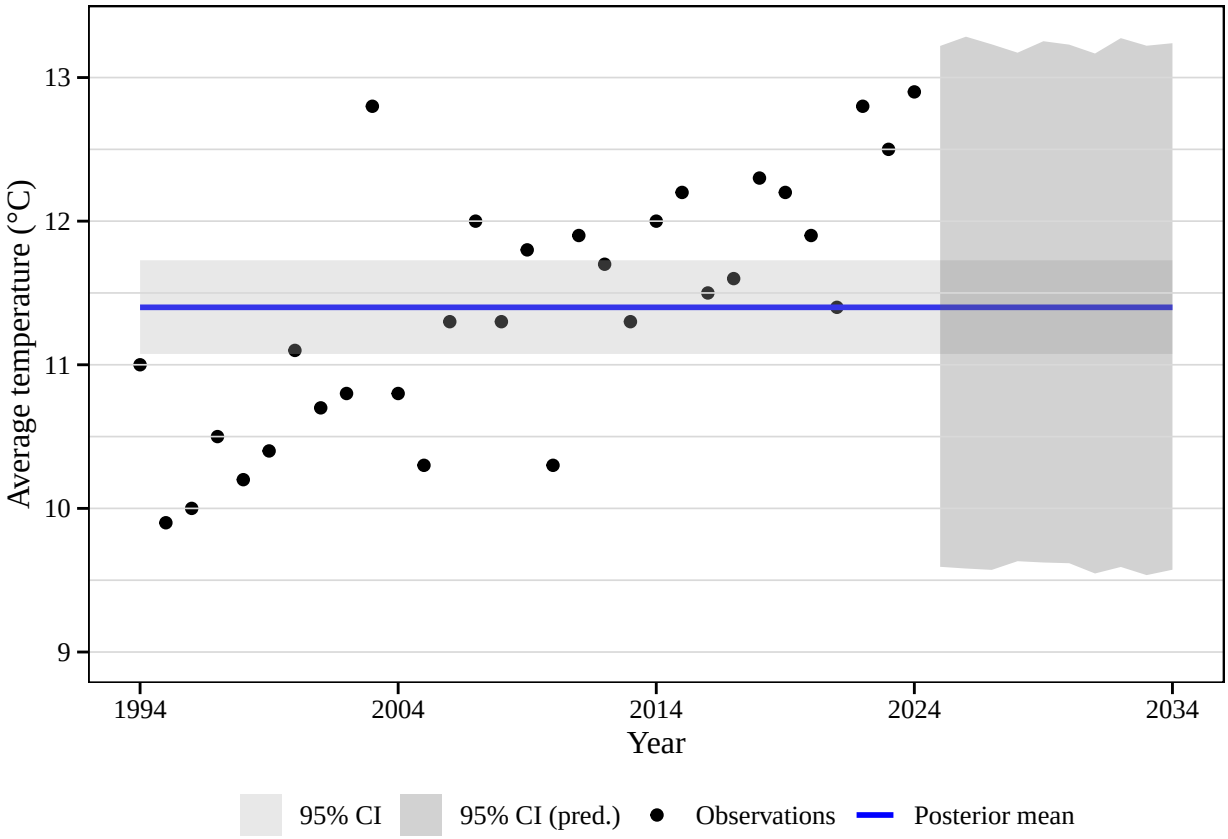



```
cor_plot_c$cor
```

```
##                T0       sigma
## T0    1.000000000 0.002891905
## sigma 0.002891905 1.000000000
```

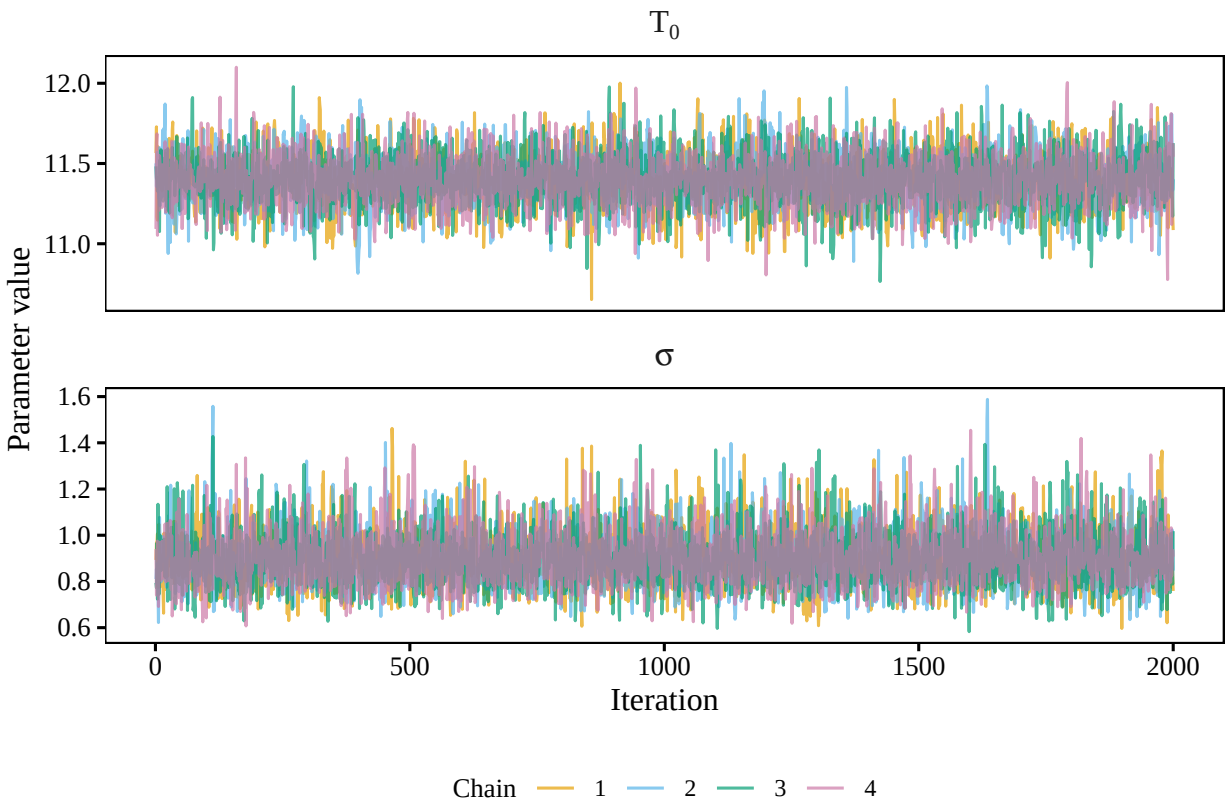
5.4 Posterior Density and Observational Fitting

```
plot_posterior(draws_df_c, my_x_stdd = stan_data_sdd$time, my_y_data = stan_data_sdd$T)
```

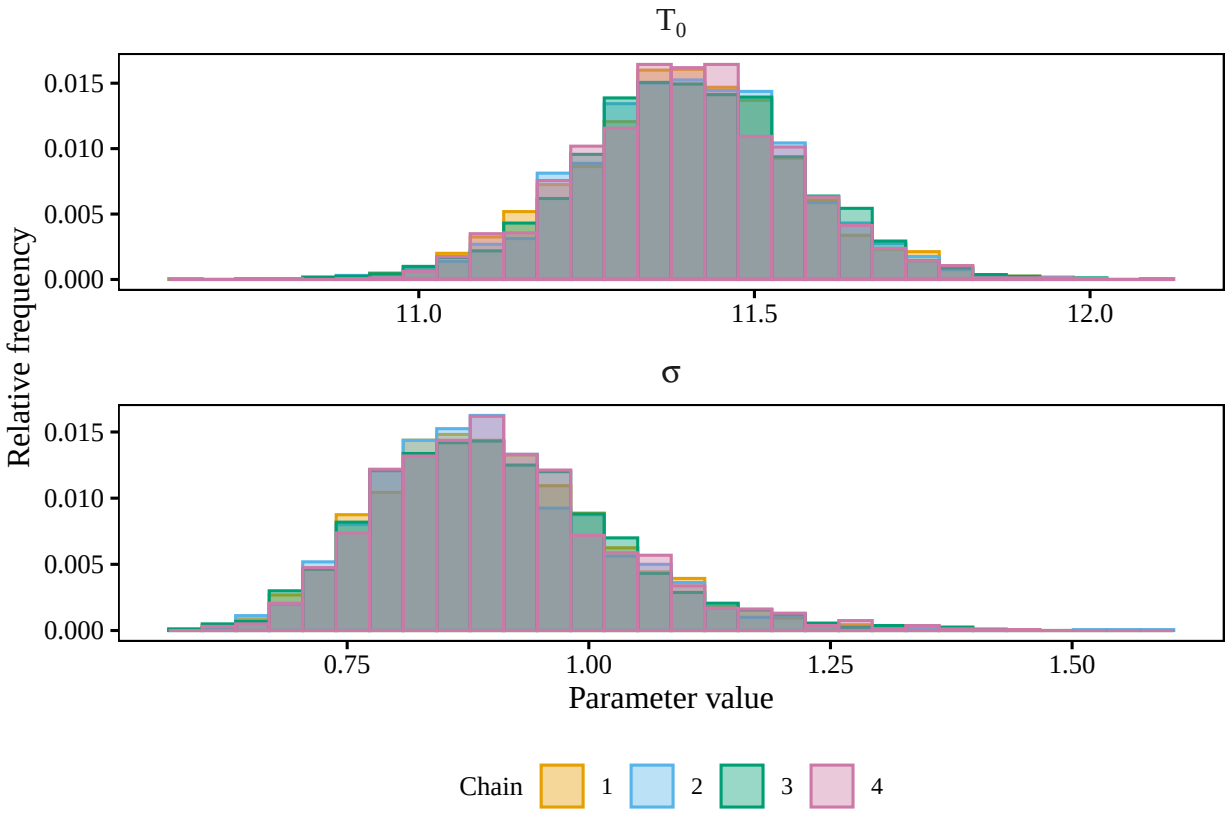


5.5 MCMC Diagnostic Plots

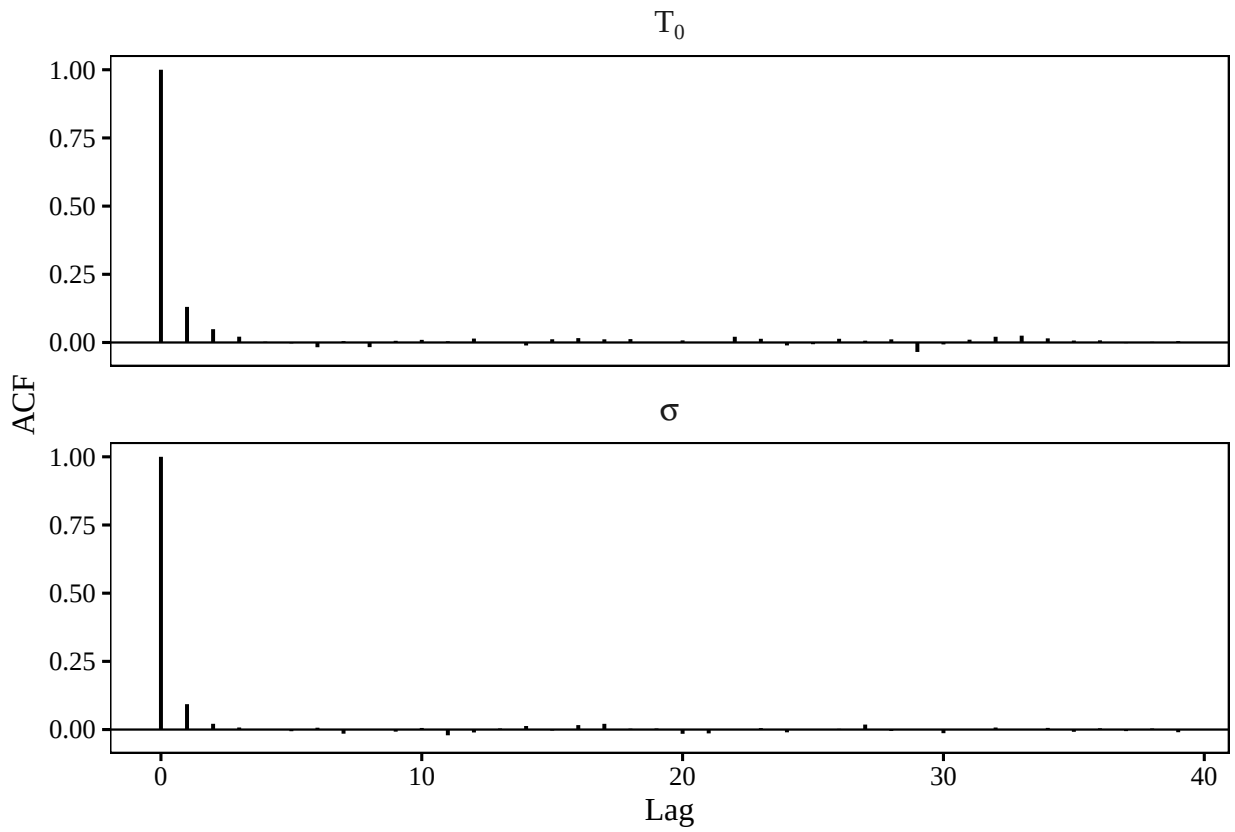
```
c_graphs <- plot_various_graphs(fit_c, c("T0", "sigma"))
print(c_graphs$trace)
```




```
print(c_graphs$density)
```




```
print(c_graphs$acf)
```



6. Model Comparison

6.1 Bayes-Factor Evaluation

6.2 Leave-One-Out Cross-Validation (LOO-CV)

```
log_lik <- fit$draws("log_lik", format = "matrix")
loo <- loo(log_lik)

c_log_lik <- fit_c$draws("log_lik", format = "matrix")
c_loo <- loo(c_log_lik)

loo_comp <- compare(loo, c_loo)
elpd_diff <- loo_comp["elpd_diff"]
se_diff   <- loo_comp["se"]

cat("LOO-CV MODEL COMPARISON\n\n")
```

```
## LOO-CV MODEL COMPARISON
```

```
cat("ELPD difference (M_Lin - M_Const):", round(elpd_diff, 3), "\n")
```

```
## ELPD difference (M_Lin - M_Const): -11.209
```

```
cat("Standard error:", round(se_diff, 3), "\n")
```

```
## Standard error: 5.944
```

7. 10-Year Posterior Predictions

...

8. Discussion

...